StudySapiens Board MCQs

Chapter 8 — Plant Physiology

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. Macronutrients are required by plants in:

(a) Only gaseous form
(b) No amounts
(c) Trace amounts only
(d) Relatively large amounts

2. Which of the following is a macronutrient?

(a) Iron
(b) Copper
(c) Zinc
(d) Nitrogen

3. Which of the following is a micronutrient?

(a) Nitrogen
(b) Calcium
(c) Molybdenum
(d) Potassium

4. Magnesium is essential because it is the core of:

(a) ATP only
(b) Cellulose
(c) Haemoglobin
(d) Chlorophyll

5. Nitrogen deficiency in plants causes:

(a) Excess flowering
(b) Root swelling only
(c) Stunted growth and leaf loss
(d) Increased chlorophyll

6. Phosphorus is a component of:

(a) Cellulose only
(b) ATP
(c) Haemoglobin
(d) Chlorophyll only

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7. Potassium is mainly involved in:

(a) Cell wall lignin
(b) DNA bases
(c) Water regulation and transport of reserve substances
(d) Chlorophyll core

8. Calcium is important for:

(a) Protein hormones
(b) Cell wall stability
(c) Oxygen transport
(d) Chlorophyll core only

9. Iron is essential for:

(a) Starch storage only
(b) Cellulose synthesis only
(c) Chlorophyll synthesis
(d) Lignin formation only

10. Insectivorous plants are still autotrophs because they:

(a) Photosynthesize
(b) Are fungi
(c) Are parasites
(d) Cannot make food

11. Pitcher plant traps insects using:

(a) Stem spines only
(b) Flowers only
(c) Modified leaves forming a pitcher
(d) Roots

12. Venus fly trap closes when prey touches:

(a) Xylem
(b) Stomata
(c) Phloem
(d) Trichomes

13. Sundew traps insects using:

(a) Pitfall pitchers
(b) Snap traps only
(c) Root nodules
(d) Sticky glandular tentacles

14. Stomata are mainly involved in:

(a) Mineral storage only
(b) Protein synthesis
(c) DNA replication
(d) Gaseous exchange and transpiration

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15. Guard cells are:

(a) Without nuclei
(b) Bean-shaped and contain chloroplasts
(c) Part of xylem
(d) Dead and lignified

16. According to the starch-sugar hypothesis, stomata open during the day because guard cells:

(a) Accumulate sugar and become turgid
(b) Lose K+ only
(c) Lose water
(d) Become flaccid

17. In the K+ influx mechanism, stomatal opening is caused by:

(a) Closure of water channels
(b) Loss of K+ from guard cells
(c) Active uptake of K+ into guard cells
(d) Abscisic acid release

18. Abscisic acid causes stomatal closure during:

(a) High sugar only
(b) Water stress/wilting
(c) High K+ only
(d) Night only

19. Blue light promotes stomatal opening by enhancing:

(a) Starch formation only
(b) Water loss only
(c) K+ uptake
(d) ABA synthesis only

20. Parenchyma supports herbaceous plants mainly through:

(a) Suberin only
(b) Lignin only
(c) Silica only
(d) Turgor pressure

21. Collenchyma provides support in:

(a) Root hairs only
(b) Young growing parts
(c) Dead woody parts only
(d) Xylem vessels only

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22. Sclerenchyma cells are usually:

(a) Living and thin-walled
(b) Photosynthetic only
(c) Without walls
(d) Dead with lignified secondary walls

23. The tonoplast is the membrane of the:

(a) Mitochondrion
(b) Nucleus
(c) Vacuole
(d) Chloroplast

24. Water potential is represented by:

(a) pH
(b) Ψw
(c) Rf
(d) pKa

25. The water potential of pure water is:

(a) Positive
(b) Infinity
(c) Negative
(d) Zero

26. Solute potential is always:

(a) Zero
(b) Equal to pressure potential
(c) Positive
(d) Negative

27. Water moves from:

(a) Lower to higher water potential
(b) Low pressure to high pressure only
(c) Equal potentials only
(d) Higher water potential to lower water potential

28. The equation for water potential is:

(a) $\Psi w = \Psi s - \Psi p$
(b) $\Psi w = \Psi s$ only
(c) $\Psi w = \Psi s + \Psi p$
(d) $\Psi w = \Psi p$ only

29. Most water and mineral uptake occurs through:

(a) Stem nodes
(b) Root hairs
(c) Root cap only
(d) Leaf veins

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30. The apoplast pathway moves water through:

(a) Cytoplasm only
(b) Vacuoles only
(c) Plasmodesmata only
(d) Cell walls

31. The symplast pathway moves water through:

(a) Cell walls only
(b) Plasmodesmata-connected cytoplasm
(c) Xylem vessels only
(d) Vacuoles only

32. The Casparian strip blocks the apoplast pathway in the:

(a) Epidermis
(b) Endodermis
(c) Cortex only
(d) Pith

33. Xylem vessels are:

(a) Dead tubular cells with perforation plates
(b) Thin-walled parenchyma only
(c) Photosynthetic cells
(d) Living cells with sieve plates

34. Tracheids differ from vessels in that they:

(a) Are living
(b) Have tapered ends and bordered pits
(c) Have perforation plates only
(d) Lack lignin

35. The TACT mechanism stands for:

(a) Turgor, Active transport, Cohesion, Tension
(b) Transpiration, Adhesion, Cohesion, Tension
(c) Transport, Absorption, Condensation, Translocation
(d) Transpiration, Active transport, Cohesion, Turgor

36. The main driving force for water movement in xylem is:

(a) Root pressure only
(b) Transpiration pull
(c) Active pumping by xylem
(d) Photosynthesis only

37. Cohesion refers to attraction between:

(a) Water molecules
(b) Solute and water
(c) Water and xylem walls
(d) Sugar and sieve tubes

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38. Adhesion refers to attraction between:

(a) Sugar molecules
(b) Water molecules only
(c) Guard cells
(d) Water and xylem walls

39. Phloem transports:

(a) Water only
(b) Organic nutrients/sugars
(c) Oxygen only
(d) Minerals only

40. Sieve tube elements are associated with:

(a) Tracheids only
(b) Root hairs only
(c) Guard cells only
(d) Companion cells

41. The pressure-flow theory was proposed by:

(a) Gregor Mendel
(b) Robert Hooke
(c) Charles Darwin
(d) Ernst Munch

42. In phloem, the source is the region where:

(a) Sugar is consumed only
(b) Minerals are absorbed
(c) Water is lost only
(d) Sugar is produced/loaded

43. In phloem, the sink is the region where:

(a) Water enters xylem only
(b) Sugar is produced only
(c) Transpiration occurs
(d) Sugar is consumed or stored

44. Sucrose is loaded into phloem at the source by:

(a) Transpiration only
(b) Active transport
(c) Diffusion only
(d) Osmosis only

45. Water enters sieve tubes at the source from the:

(a) Xylem
(b) Stomata
(c) Epidermis only
(d) Phloem fibres only

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46. The pressure-flow theory explains movement from source to sink due to:

(a) Active pumping along the whole tube
(b) Diffusion only
(c) Gravity only
(d) A pressure gradient

47. Meristems are regions of:

(a) Dead cells
(b) Active cell division
(c) Food storage only
(d) Permanent differentiation only

48. Apical meristems are responsible for:

(a) Bark formation only
(b) Secondary growth/girth
(c) Primary growth/length
(d) Fruit ripening only

49. Lateral meristems are responsible for:

(a) Transpiration only
(b) Length only
(c) Flowering only
(d) Secondary growth/thickness

50. Intercalary meristems are common in:

(a) Ferns only
(b) Grasses
(c) Woody trees only
(d) Mosses only

51. Vascular cambium produces secondary xylem toward the:

(a) Epidermis only
(b) Outside
(c) Inside
(d) Pith only

52. Vascular cambium produces secondary phloem toward the:

(a) Root cap
(b) Outside
(c) Xylem only
(d) Inside

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53. Cork cambium produces:

(a) Root hairs
(b) Cork/bark
(c) Secondary xylem only
(d) Secondary phloem only

54. Annual rings are formed by seasonal activity of the:

(a) Epidermis
(b) Cambium
(c) Root cap
(d) Stomata

55. Spring wood is usually:

(a) Lighter with wider vessels
(b) Absent
(c) Only phloem
(d) Darker with narrow vessels

56. Dendrochronology is used to determine:

(a) Soil pH
(b) Transpiration rate only
(c) Leaf area only
(d) Tree age and past climate

57. Auxins promote:

(a) Seed dormancy only
(b) Cell elongation and apical dominance
(c) Stomatal closure only
(d) Fruit ripening only

58. Cytokinins mainly promote:

(a) Dormancy only
(b) Cell division
(c) Abscission only
(d) Stomatal closure only

59. Gibberellins promote:

(a) Stomatal closure only
(b) Leaf fall only
(c) Dormancy only
(d) Stem elongation and seed germination

60. Abscisic acid is associated with:

(a) Stress response and dormancy
(b) Stem elongation only
(c) Cell division only
(d) Fruit ripening only

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61. Ethylene is mainly associated with:

(a) Root initiation only
(b) Fruit ripening
(c) Cell division only
(d) Stomatal opening only

62. A hypotonic solution causes plant cells to:

(a) Remain unchanged
(b) Die immediately
(c) Gain water and become turgid
(d) Lose water and plasmolyse

63. A hypertonic solution causes plant cells to undergo:

(a) Turgor increase
(b) Deplasmolysis
(c) Cell division
(d) Plasmolysis

64. Hydrophytes are adapted to:

(a) Deserts only
(b) Aquatic environments
(c) Saline soils only
(d) Cold mountains only

65. Xerophytes are adapted to:

(a) Deep shade only
(b) Aquatic environments only
(c) Saline soils only
(d) Dry environments

66. Halophytes are adapted to:

(a) Aquatic submerged only
(b) Saline environments
(c) Freshwater only
(d) Deserts only

67. Mesophytes live in:

(a) Extreme deserts only
(b) Saline marshes only
(c) Moderate environments
(d) Deep oceans only

68. Heat-shock proteins help plants by:

(a) Preventing protein denaturation at high temperature
(b) Closing stomata only
(c) Forming ice crystals
(d) Increasing transpiration only

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69. Plants adapt to cold by increasing membrane:

(a) Lignin only
(b) Cellulose only
(c) Unsaturated fatty acids
(d) Saturated fatty acids only

70. Intracellular ice formation is dangerous because it:

(a) Opens stomata
(b) Increases photosynthesis
(c) Perforates membranes and organelles
(d) Produces ATP

71. Phototropism is growth response to:

(a) Gravity
(b) Light
(c) Chemicals
(d) Touch

72. Geotropism is growth response to:

(a) Gravity
(b) Water only
(c) Touch
(d) Light

73. Thigmotropism is growth response to:

(a) Chemicals only
(b) Touch
(c) Gravity
(d) Light

74. Chemotropism is growth response to:

(a) Chemicals
(b) Gravity
(c) Touch only
(d) Light

75. Photoperiodism is the response of plants to:

(a) Length of day and night
(b) Wind only
(c) Soil pH only
(d) Temperature only

76. Short-day plants actually require:

(a) Short nights only
(b) High temperature only
(c) Long uninterrupted nights
(d) Continuous light

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77. Long-day plants flower when:

(a) Nights are short or interrupted
(b) Temperature is low only
(c) Days are short only
(d) Nights are long and uninterrupted

78. Phytochrome exists in two forms called:

(a) P660 and P730
(b) P680 and P700
(c) Pr and Pg only
(d) Pfr and Pgr only

79. Red light converts phytochrome from:

(a) P660 to P730
(b) P700 to P680
(c) P730 to P660
(d) P680 to P700

80. Far-red light converts phytochrome from:

(a) P660 to P730
(b) P680 to P700
(c) P730 to P660
(d) P700 to P680

81. A flash of light during the night prevents flowering in:

(a) All plants equally
(b) Long-day plants only
(c) Short-day plants
(d) Day-neutral plants only

82. Vernalisation is induction of flowering by:

(a) Low temperature
(b) Long days only
(c) High temperature only
(d) Touch only

83. The site of vernalisation perception is the:

(a) Shoot apex or embryo
(b) Root hair only
(c) Stomata only
(d) Leaf only

84. Vernalisation is especially important for:

(a) Biennial plants
(b) Aquatic algae only
(c) Annual weeds only
(d) Fungi only

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85. The hormone-like substance associated with vernalisation is:

(a) Vernalin
(b) Abscisic acid only
(c) Auxin only
(d) Ethylene only

86. Bark includes tissues outside the:

(a) Epidermis only
(b) Pith
(c) Vascular cambium
(d) Root cap

87. The zone of cell elongation is characterized by:

(a) Vacuolated enlarging cells
(b) Only dead cells
(c) Only differentiated cells
(d) Only dividing cells

88. Primary growth results in increase of:

(a) Bark only
(b) Girth only
(c) Fruit size only
(d) Length

89. Secondary growth results in increase of:

(a) Length only
(b) Thickness/girth
(c) Leaf number only
(d) Flower number only

90. Which pathway is negligible in water movement through roots?

(a) Apoplast pathway
(b) Vacuolar pathway
(c) Symplast pathway
(d) Xylem pathway

91. Companion cells supply sieve tube elements with:

(a) Lignin only
(b) ATP and proteins
(c) Water only
(d) DNA only

92. Translocation of food occurs mainly in the:

(a) Epidermis
(b) Xylem
(c) Cuticle
(d) Phloem

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93. The ascent of sap occurs mainly in the:

(a) Pith only
(b) Cortex only
(c) Xylem
(d) Phloem

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ANSWER KEY WITH BRIEF REASONS

1. (d) Relatively large amounts — Macronutrients are needed in larger quantities.

2. (d) Nitrogen — N is required in large amounts.

3. (c) Molybdenum — Mo is needed in trace amounts.

4. (d) Chlorophyll — Mg sits at the centre of chlorophyll.

5. (c) Stunted growth and leaf loss — N is needed for proteins and growth.

6. (b) ATP — ATP contains phosphate.

7. (c) Water regulation and transport of reserve substances — K regulates osmotic/stomatal functions.

8. (b) Cell wall stability — Ca stabilizes walls/membranes.

9. (c) Chlorophyll synthesis — Fe is needed for chlorophyll formation.

10. (a) Photosynthesize — They supplement nitrogen by trapping insects.

11. (c) Modified leaves forming a pitcher — Leaves form a pitfall trap.

12. (d) Trichomes — Trigger hairs cause rapid closure.

13. (d) Sticky glandular tentacles — Dew-like mucilage traps prey.

14. (d) Gaseous exchange and transpiration — Stomata regulate gas/water vapour movement.

15. (b) Bean-shaped and contain chloroplasts — Guard cells control stomatal aperture.

16. (a) Accumulate sugar and become turgid — Sugar lowers water potential, water enters.

17. (c) Active uptake of K+ into guard cells — K+ lowers osmotic potential, water enters.

18. (b) Water stress/wilting — ABA stops K+ pumping under stress.

19. (c) K+ uptake — Blue light stimulates proton pumps/K+ uptake.

20. (d) Turgor pressure — Water-filled parenchyma cells provide turgidity.

21. (b) Young growing parts — Collenchyma is flexible support.

22. (d) Dead with lignified secondary walls — Sclerenchyma gives rigid support.

48. (c) Primary growth/length — They occur at root/shoot tips.

49. (d) Secondary growth/thickness — Vascular/cork cambium increase girth.

50. (b) Grasses — They occur at internode bases.

51. (c) Inside — Inner derivatives become wood.

52. (b) Outside — Outer derivatives become inner bark.

53. (b) Cork/bark — It forms protective periderm.

54. (b) Cambium — Spring/autumn wood create rings.

55. (a) Lighter with wider vessels — Favourable spring conditions produce wide cells.

56. (d) Tree age and past climate — Ring patterns record age/environment.

57. (b) Cell elongation and apical dominance — IAA regulates growth.

58. (b) Cell division — They stimulate cytokinesis.

59. (d) Stem elongation and seed germination — GA stimulates growth/germination.

60. (a) Stress response and dormancy — ABA helps tolerate stress.

61. (b) Fruit ripening — Ethylene ripens fruit.

62. (c) Gain water and become turgid — Water enters the cell.

63. (d) Plasmolysis — Water leaves the protoplast.

64. (b) Aquatic environments — They live in water.

65. (d) Dry environments — They conserve water.

66. (b) Saline environments — They tolerate salt.

67. (c) Moderate environments — They need neither extreme adaptation.

68. (a) Preventing protein denaturation at high temperature — They protect proteins under heat stress.

69. (c) Unsaturated fatty acids — Unsaturated lipids maintain fluidity.

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23. (c) Vacuole — Tonoplast surrounds the vacuole.
24. (b) Ψw — Psi denotes water potential.
25. (d) Zero — Pure water is the reference at zero.
26. (d) Negative — Solutes lower water potential.
27. (d) Higher water potential to lower water potential — Water follows its potential gradient.
28. (c) Ψw = Ψs + Ψp — Solute plus pressure potential.
29. (b) Root hairs — Root hairs increase surface area.
30. (d) Cell walls — Apoplast is the non-living wall pathway.
31. (b) Plasmodesmata-connected cytoplasm — Symplast is the living continuum.
32. (b) Endodermis — It forces selective uptake into symplast.
33. (a) Dead tubular cells with perforation plates — Vessels conduct water efficiently.
34. (b) Have tapered ends and bordered pits — Tracheids are less specialized conduits.
35. (b) Transpiration, Adhesion, Cohesion, Tension — It explains ascent of sap.
36. (b) Transpiration pull — Evaporation creates tension.
37. (a) Water molecules — Hydrogen bonding keeps the column continuous.
38. (d) Water and xylem walls — Adhesion helps resist gravity.
39. (b) Organic nutrients/sugars — Phloem translocates food.
40. (d) Companion cells — Companion cells support sieve tubes.
41. (d) Ernst Munch — Munch proposed mass flow.
42. (d) Sugar is produced/loaded — Leaves are typical sources.
43. (d) Sugar is consumed or stored — Roots/fruits are sinks.
44. (b) Active transport — Loading requires energy.
45. (a) Xylem — Osmosis draws water from nearby xylem.
46. (d) A pressure gradient — Loading/unloading creates pressure difference.
47. (b) Active cell division — Meristematic cells divide continuously.
70. (c) Perforates membranes and organelles — Ice crystals damage cells.
71. (b) Light — Shoots bend toward light.
72. (a) Gravity — Roots grow downward.
73. (b) Touch — Climbing tendrils coil on contact.
74. (a) Chemicals — Fungal hyphae/pollen tubes respond to chemicals.
75. (a) Length of day and night — It regulates flowering.
76. (c) Long uninterrupted nights — Dark period length is critical.
77. (a) Nights are short or interrupted — They need short critical night.
78. (a) P660 and P730 — Red/far-red absorbing forms.
79. (a) P660 to P730 — Red light activates the P730 form.
80. (c) P730 to P660 — Far-red reverses the active form.
81. (c) Short-day plants — It interrupts the long dark requirement.
82. (a) Low temperature — Cold treatment promotes flowering.
83. (a) Shoot apex or embryo — Cold is sensed by meristematic regions.
84. (a) Biennial plants — Many biennials require cold.
85. (a) Vernalin — Vernalin is linked to cold-induced flowering.
86. (c) Vascular cambium — Secondary phloem/cork form bark.
87. (a) Vacuolated enlarging cells — Cells increase in volume.
88. (d) Length — Apical meristems extend the plant.
89. (b) Thickness/girth — Lateral meristems add girth.
90. (b) Vacuolar pathway — Vacuolar route contributes little.
91. (b) ATP and proteins — They support sieve-tube metabolism.
92. (d) Phloem — Phloem carries sugars.
93. (c) Xylem — Xylem conducts water upward.

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